

Argand diagrams

Key Points

For a complex number

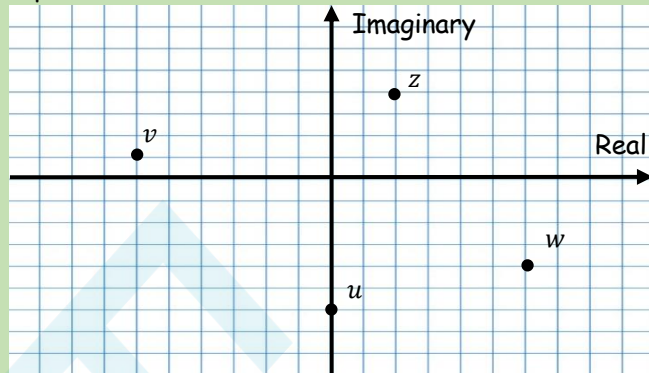
$$z = x + yi$$

The plot of (x, y) is an argand diagram with the x-axis representing the real value and the y-axis representing the imaginary value.

Using a complex number's modulus and argument you can plot a point on the argand diagram by a distance and

Basic Skills

Q1. Using the argand diagram below, write the values of the complex number



a) z b) w c) u d) v

Competency Skills

Q2. Plot the following complex numbers on an argand diagram

a) $2 + 3i$ b) $4 - 5i$ c) $5i$ d) -3 e) $-4 + i$

Q3. a) Show that $\frac{4i}{3-i} = \frac{12i-4}{10}$

b) Hence plot the complex number $z = \frac{4i}{3-i}$ on an argand diagram

Q4. For the complex numbers, $z = 2 + i$ and $w = 3i - 4$ plot on an argand diagram

a) z b) w c) $z - w$ d) $3z + 2w$

e) z^* f) $z^* - w^*$ g) $\frac{z}{w}$ h) $\frac{w}{z}$

Advanced Skills

Q5. a) Plot the complex number $z = 3 + 4i$ on an argand diagram

b) Find the distance the complex number is from the origin

c) Find the angle that the complex number makes with the positive real axis

d) Hence suggest a non-cartesian way to describe complex numbers

Q6. A student is plotting solutions to quadratic equations on an argand diagram

a) Explain what they should expect to see if the discriminant is greater than zero

b) After plotting a solution to quadratic the student states, "One solution is just reflected in the y-axis to get other", state the value of the discriminant for the quadratic

Extension

Q7. Generalise the approach in Q5. to describe a general complex number $z = x + yi$ in terms of its distance from the origin and its angle

Hint: Draw a triangle